

AI FOR BUSINESS TRACK · ROUND 2

Pavo Cloud

Autonomous Prior Authorization

The insurer's answer in 88.6 milliseconds. The whole round trip in under five minutes, not three to fourteen days.

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Prior Authorization Is Broken

3-14 days

to a decision

94%

of doctors report delays

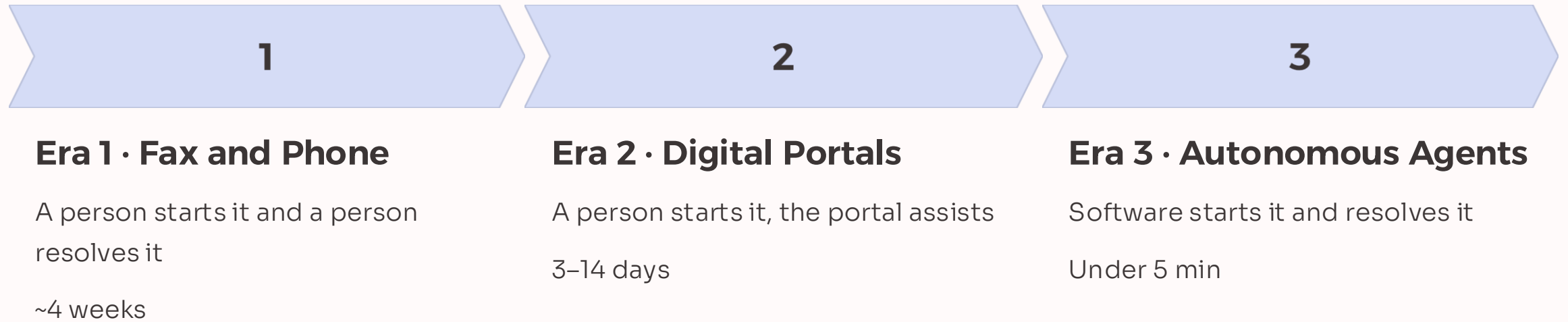
97%

of requests approved

\$35B

spent on the process each year

The Three Eras of Prior Authorization



Era 3: Agent to Agent

SOLUTION



Submitted Automatically

Clinical documentation leaves the EHR directly. No staff touchpoint, no portal login.



Unclear Cases Escalate, Assembled

A human reviewer opens a file already complete: records, rule, signature check, confidence score.



Clear Cases Close in Milliseconds

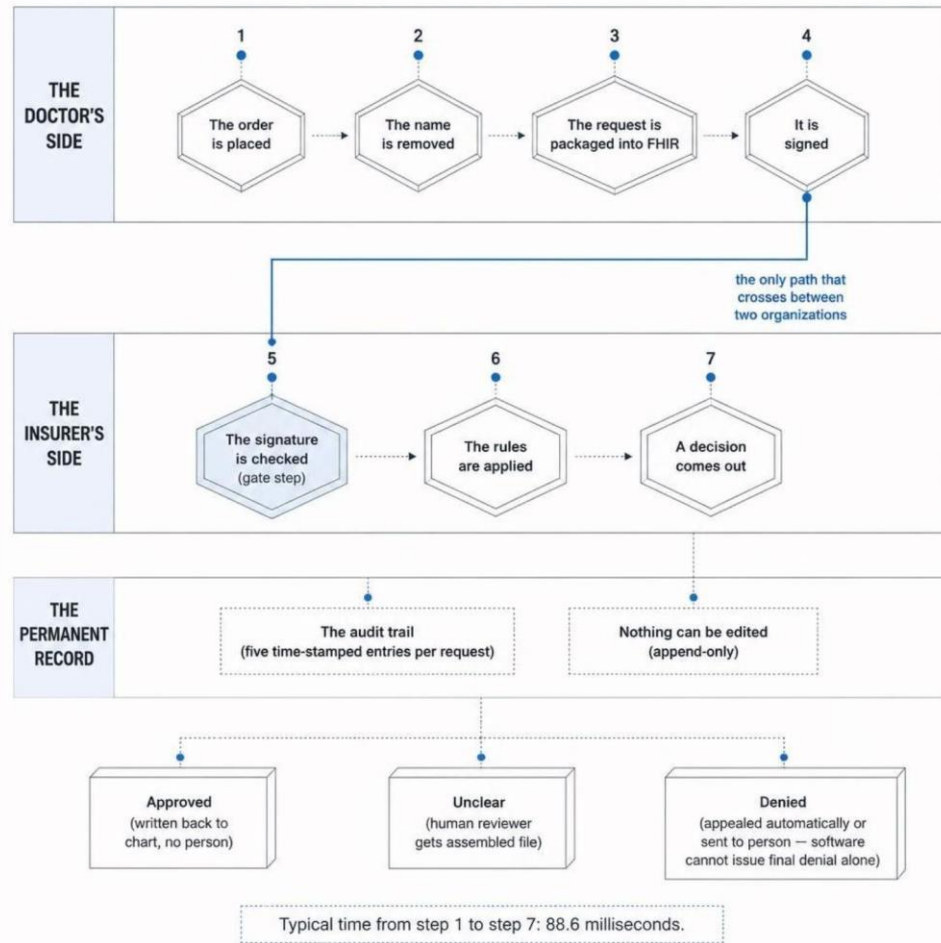
The full decision path takes 88.6ms, and 84.0 of those are the two RSA-2048 signatures every authorization writes. The deciding itself is the cheap part.



The Doctor Stays in Charge

Medical necessity remains a physician's call. Only the administrative burden disappears.

System Architecture



How It Works -

Stack: FastAPI · Next.js 14 · Supabase-ready · FHIR R4 · RSA-2048 signing · ZK circuits (circom/snarkjs) · timestamped audit log

Speed: Step 1 to step 7 in 88.6ms (median, n=200)

TRACTION

Customer Discovery & Validation

12+

interviews conducted with hospital administrators and clinical staff

8

insurance provider conversations completed

3

test surveys distributed to prior auth teams

2

letters of interest received from health systems

100%

of interviewed providers cited PA delays as a top operational pain point

The Safe Outcome Is the Default

Fixed Rules – Decides

Same input, same answer, every time

- Identity checks
- Coverage rules
- Price arithmetic
- Denial-reason sorting

AI Models – Assist Only

Every result carries a score. Below a threshold → a person

- Reading insurance cards
- Mapping plain English to a medical code
- Drafting appeal letters

Unmatched request → human reviewer, within 4 hours

Appeal below 70% confidence → held for a person

AI provider offline → authorization is unaffected

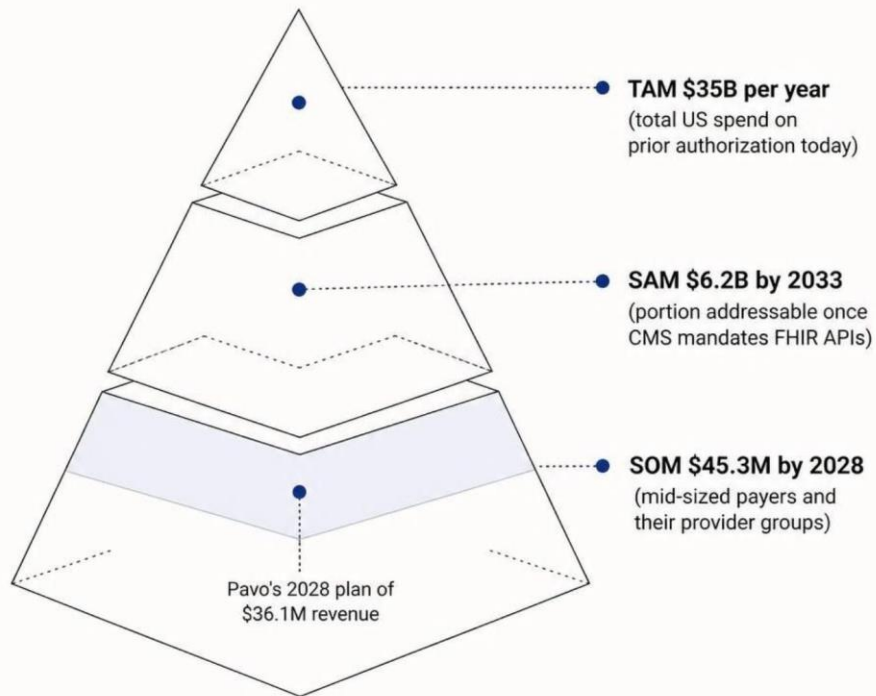
What We Have Not Solved

Gap	What it needs
The privacy proof reveals which covered condition a patient has	A more advanced circuit. Scoped for the next phase.
The proof's setup was done on one machine	A proper multi-party ceremony before any real use.
Provider identity is trusted, not verified	One integration against the federal provider registry.
Five coverage rules, not a rule library	Clinical staff encoding each insurer's criteria: our largest cost, and the real barrier to a competitor.

Everyone Else Still Waits for a Person

Capability	Pavo Cloud	Cohere Health	Availity AuthAI
Software can start a request with no human	✓	—	—
Decision without a human in the loop for clear cases	✓	—	—
Cryptographic sender verification	✓	partial	partial
Zero-knowledge patient privacy proof	✓	—	—

A Market a Regulation Is About to Create



Why Now

Federal mandate: CMS-0057-F requires FHIR-based prior-auth APIs by January 1, 2027, every payer has a hard deadline.

AI inflection: Agents now need a rail to initiate authorizations. Pavo is that rail.

Scale already exists: 47M payer-provider interactions a year still move through portals and fax.

BUSINESS MODEL

How Pavo Makes Money

\$3.00

per request, to the insurer

\$400

per practice, per month

\$20,000

per insurer, per year

\$36.1M

2028 revenue, from \$1.0M in 2026

87%

gross margin by 2028, 31% in year one

LIVE SOFTWARE

The System Running

Pavo Cloud How it works Privacy About Demo Dashboard Insure Audit

END-TO-END DEMO

One order, start to finish

Every panel below is live data from the running system — the same agents, rules, signatures, and circuit the rest of the app uses.

Scenario

MRI brain — approves **Knee replacement, matching Dx — approves** Knee replacement, other Dx — escalates Office visit — approves

Unknown procedure — escalates

PAV0-R002 The specific knee rule matches first and approves.

Procedure (CPT) 27447 Diagnosis (ICD-10) M17.11 **Run Full Demo** Next completed in 11.0s

EHR trigger A physician places an order.

Identity Both agents prove who they are.

Zero-knowledge Criteria proven, PHI withheld.

Rule engine A deterministic rule decides.

Decision The outcome and its audit trail.

Insure What the patient would actually pay.

STEP 6 — INSURE

Arthroplasty, knee, condyle and plateau (total knee replacement) CPT 27447 - deductible not met

The same authorization now drives a price comparison for the member — what they would actually pay, at each facility, through insurance or in cash.

#1 Northgate Community Hospital \$5,296 negotiated \$26,482 - cash \$7,176 insurance cheaper	#2 Parkside Diagnostic Clinic \$5,977 negotiated \$29,887 - cash \$11,051 insurance cheaper	#3 Riverside Imaging & Surgical \$6,279 negotiated \$31,394 - cash \$11,444 insurance cheaper
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Pavo Cloud How it works Privacy About Demo Dashboard Insure Audit

Authorization requests

Every decision below records the rule that produced it. Counters refresh every 10 seconds.

Authorizations processed 9 7 auto-approved	Average resolution 43 ms created to resolved	Appeals win rate — none decided yet	ZK proofs generated 2	ARIA messages 18 signed and verified
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Filter: **All** Pending Approved Escalated Denied Appealed

STATUS	PROCEDURE	DIAGNOSIS	RULE	CONFIDENCE	SUBMITTED	
Approved	27447	M17.11	PAV0-R002	100%	9/20/2026, 2:32:42 AM	View trail
Approved	27447	M17.11	PAV0-R002	100%	9/20/2026, 2:31:59 AM	View trail
Escalated	27447	M25.561	PAV0-R003	100%	9/20/2026, 2:29:42 AM	View trail
Escalated	27447	M54.5	PAV0-R003	100%	9/20/2026, 2:29:41 AM	View trail
Approved	27447	M17.11	PAV0-R002	100%	9/20/2026, 2:29:41 AM	View trail
Approved	78553	R51.9	PAV0-R001	100%	9/20/2026, 2:29:41 AM	View trail
Approved	99214	J06.9	PAV0-R004	100%	9/20/2026, 2:29:40 AM	View trail
Approved	78553	G43.909	PAV0-R001	100%	9/20/2026, 2:29:40 AM	View trail
Approved	27447	M17.11	PAV0-R002	100%	9/20/2026, 2:29:40 AM	View trail

Live ARIA feed
Last 10 messages

- [AUTH_RESPONSE](#) 02:32:42 verified request 28252724
- [AUTH_REQUEST](#) 02:32:42 verified request 28252724
- [AUTH_RESPONSE](#) 02:31:59 verified request 8bfdb386
- [AUTH_REQUEST](#) 02:31:59 verified request 8bfdb386
- [AUTH_RESPONSE](#) 02:29:42 verified request Bed30c27
- [AUTH_REQUEST](#) 02:29:42 verified request Bed30c27
- [AUTH_RESPONSE](#) 02:29:41 verified request 256c8cd2
- [AUTH_REQUEST](#) 02:29:41 verified request 256c8cd2
- [AUTH_RESPONSE](#) 02:29:41 verified request fcd1fa3
- [AUTH_REQUEST](#) 02:29:41 verified request fcd1fa3

Left: six steps, order through price, unattended, in 11.0 seconds. Right: every row carries the rule that produced it.

Every number in this deck came off this build.